Physics Guided Remote Sensing Image Synthesis Network for Ship Detection

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***Abstract*— Automatic detection and localization of objects in remote sensing images are of great significance for remote sensing systems. Existing frameworks usually train an object detection network using collected remote sensing images. However, these models usually perform poorly due to the lack of large-scale training datasets, which is often the case for special remote sensing scenarios, e.g., the detection of ships in the open sea. Although image synthesis is a common strategy to alleviate the issue of data insufficiency, the trained model still performs poorly when being tested on real-world scenes. Aimed at this, a novel sensor-related image synthesis framework, dubbed as remote sensing-image synthesis pipeline (RS-ISP), is developed to address the lack of on-orbit remote sensing images. Specifically, our RS-ISP introduces two novel designs to ensure the distribution consistency between the generated images and the real images: 1) the first is a novel pipeline for modeling the physical process of noise production during image capture using specific sensors and 2) the second is the design of a detection-oriented image harmonization model. Similar to the existing design, our model first produces coarse synthetic images by copy–paste operation, on which the proposed harmonization process is used to reduce the variation in the pasted foreground and background. By incorporating these two designs into a unified framework, our RS-ISP is designed and used to produce large-scale synthetic images used to train the object detection model for detecting ships in remote sensing images. Comparative experiments demonstrated that RS-ISP increased the AP@.50 from 0.148 to 0.498 for the ship detection task. Code will be publicly available.**

***Index Terms*— Image harmonization, image synthesis, physical noise simulation, remote sensing, ship detection.**

I. Introduction

REMOTE sensing is an important means of large-scale information acquisition and monitoring in various fields, such as agriculture, ocean observation, and disaster detection. To process massive remote sensing data, deep learning methods that can automatically extract different levels of features in images have become the state-of-the-art solution for remote sensing tasks. As a data-driven method, deep learning requires a massive amount of labeled data to train networks. However, it is very difficult and expensive to acquire and label a large-scale dataset in the remote sensing field, especially the on-orbit image dataset for ship object detection. The feature distribution of remote sensing images has a strong correlation with its imaging conditions and sensor hardware. In addition, due to the high cost and irreplaceability of remote sensing sensors, sensor-related datasets and parameters must be provided for remote sensing algorithms. Remote sensing image synthesis, particularly for certain sensors and tasks, is a practical approach that can be used to overcome the difficulties listed above.

To make up for the lack of on-orbit remote sensing image data, image synthesis methods need to generate images that are as realistic as possible while satisfying the sensor characteristics. However, many issues can result in unrealistic synthetic images. There are two sorts of issues that cause unrealistic images: semantic problems and structural problems. Semantic issues include inappropriate positions, such as a cat in the sky, and structural issues typically involve irrational sizes, tone variations between the foreground and the background, and unnatural boundaries. Traditional image synthesis methods typically generate synthetic data by constructing a sample library and randomly combining the samples in the sample library through copy–paste operations. Copy–paste operation is a simple way to fuse the target and the background, but it will lead to unnatural edges and inconsistency between the target and the background. The unnaturalness of the edges, as well as the incongruity of the foreground and background, can cause a distribution shift between the synthetic and actual data, making it hard to use it directly to alleviate the problem of insufficient data. Many efforts have been made to solve the inconsistency of the synthetic image [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15]. Approaches based on generative adversarial networks (GANs) have a good capacity to produce realistic enough images that are extremely similar to real images [11]. However, it still generates images containing incorrect semantic information and structural information, and it is difficult to explicitly control its generation results. Furthermore, the synthetic images generated by both image blending and GANs lack the physical information in the

Manuscript received 17 August 2022; revised 9 December 2022 and 6 February 2023; accepted 15 February 2023. Date of publication 23 February 2023; date of current version 7 March 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 62102069, Grant U20B2063, and Grant 62220106008; in part by the Sichuan Science and Technology Program under Grant 2022YFG0032; and in part by the China Academy of Space Technology (CAST) Innovation Program. *(Weichang Zhang and Rui Zhang are co-first authors.) (Corresponding authors: Guoqing Wang; Wei Li.)*

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Digital Object Identifier 10.1109/TGRS.2023.3248106

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imaging link, such as sensor noises and lighting conditions, which is especially vital for remote sensing images. Although these works have made some progress in resolving the distribution shift issue with synthetic data, it is still challenging to use their synthetic results to compensate for the data insufficiency.

In this article, we proposed a novel image synthesis framework called remote sensing-image synthesis pipeline (RS-ISP) for remote sensing applications to address the issue of insufficient on-orbit data in ship detection, by introducing a deep-learning-based image harmonization method to eliminate the illumination difference between the foreground and the background of the synthetic image, and sensor noise to further increasing the realism of the synthetic image. Our proposed RS-ISP pipeline consists of three stages: coarse image synthesis stage, image harmonization stage, and noise simulation stage.

In the first stage which is coarse image synthesis, we fuse the ship images transformed from the 3-D models as the foreground with the real remote sensing images of GaoFen-2 as the background by a simple copy–paste operation, and we take these images as the coarse synthesis images. In this stage, we make adjustments to the foreground image size based on sensor characteristics while imaging the background image, to ensure that an image with an unreasonable foreground size will not be generated. Coarse synthesis images generated in the first stage have a certain inconsistency between the foreground and the background. The image harmonization stage is designed to reduce these inconsistencies, and we construct a detection-oriented image harmonization network based on DoveNet [16] and YOLOv5 [17] to alleviate the domain gap between the foreground and the background of the coarse synthesis images. In the last stage, the noise simulation stage, we first analyze the sources of the noise during image capture by taking the charge-coupled device (CCD) as an example, and then simplify the noise composition and establish a distribution model of image noise used to add to the harmonized composite images to generate more realistic images. We validate the effectiveness of our image synthesis pipeline by cross-domain testing of multiple object detection models. The cross-domain test results can be used as a measurement to evaluate the realism of the synthetic data and the distribution similarity of synthetic data to real data. The experimental results show that our RS-ISP framework can generate images that are comparable to real images. The experimental results further demonstrate that our noise distribution model and detection-oriented image harmonization method could significantly enhance the performance of object detection models trained on synthetic data on real data, even when the real data contain significant noise.

Our main contributions are summarized as follows.

1) We propose a detection-oriented image harmonization method to reduce the difference between the foreground and the background of the coarse synthesis images. We use a joint training strategy to enable the generator in the image harmonization network to generate results that take into account both the visual effects and downstream tasks.

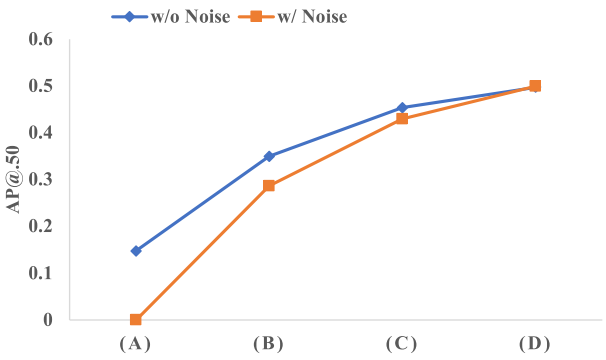


Fig. 1. Detection results on real data by testing YOLOv5s trained on synthesis images obtained with four different settings. The blue (orange) line represents the result of the real data without (with) our proposed physical noise simulation. (a) Coarse synthetic data that are obtained by directly putting the foreground into the background without any process. (b) Synthetic data that are harmonized by the image harmonization network. (c) Synthetic data that are harmonized by our proposed detection-oriented image harmonization network. While (d) is adding physically simulated noise to (c). The results show a gradually increasing trend, which proves the effectiveness of our RS-ISP.

2) We analyze the source and distribution of sensor noise during the image capture process and design a simplified noise distribution model to simulate sensor noise.
3) Combining the above two points, we propose an image synthesis framework called RS-ISP for the task of on-orbit remote sensing image object detection to alleviate the problem of insufficient on-orbit remote sensing images.

II. Related Work

A. Image Composition

Deep learning is essentially a data-driven method, and deep neural networks need a large amount of rich data to train the model to achieve a good result. However, in many application scenarios, high-quality and rich data usually require expensive manpower and resources to obtain. Data synthesis can effectively alleviate the problem of insufficient data volume and richness. Ghiasi et al. [2] and Dwibedi et al. [3] pasted the foreground and background together, but there is a certain inconsistency between the foreground and the background of the generated image and artifacts in the boundary. Dwibedi et al. [3], Pérez et al. [4], and Zhang et al. [5] tried to eliminate this inconsistency by Poisson blending and lighting variations, but it is still unable to completely eliminate boundary artifacts. Hinterstoisser et al. [6], Johnson-Roberson et al. [7], Alhaija et al. [8], and Georgakis et al. [9] used computer graphics to construct real scenes and used rendering to obtain synthetic data. These methods can achieve very good results in image quality and fidelity, but usually require expensive manpower and GPU resources to build a realistic rendering scene that is sufficiently realistic. Richter et al. [10] directly used video games as rendering scenes, but the data generated by this method usually have large domain differences between real data and are difficult to apply directly. Shamsolmoali et al. [11], Goodfellow et al. [12], Ma et al. [13],

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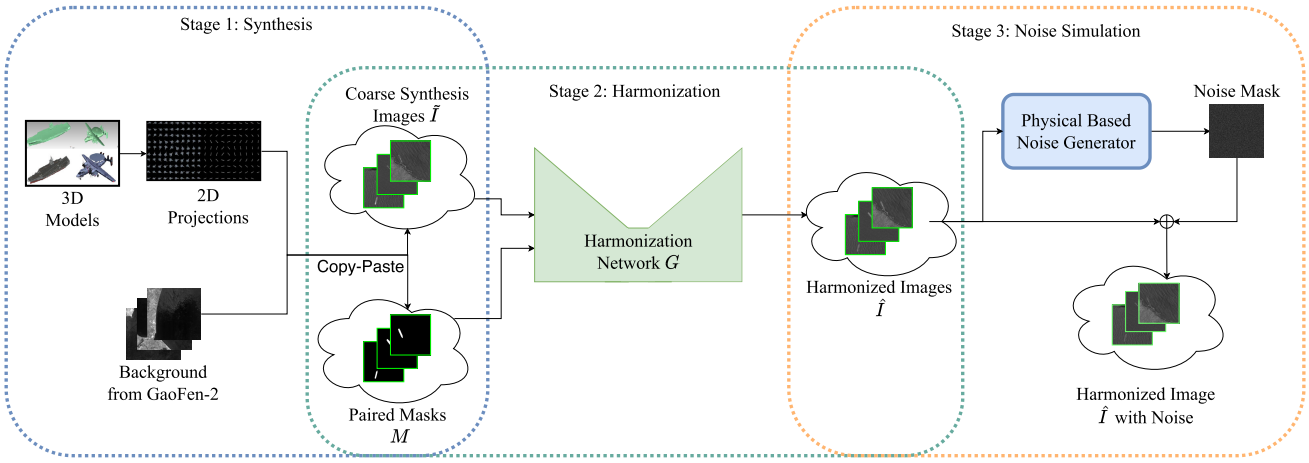


Fig. 2. Illustration of our proposed RS-ISP. Specifically, a coarse synthetic image dataset is constructed using 3-D models and backgrounds from the GaoFen-2 [1] dataset in the first stage. Then in the second stage, a detection-oriented image harmonization network is designed to reduce the discrepancy between the foreground and the background. Followed by this, we add noise to the synthetic images based on the sensor noise model to further improve the realism of the synthetic image.

Deng et al. [14], and Baier et al. [15] used GAN which has achieved extraordinary results in the field of image generation. The generated results can be very close to the distribution of real data, but usually cannot explicitly control the specific fine-grained information of the generated images.

B. Image Harmonization

Image harmonization aims to make the composite foreground compatible with the background. Traditional methods mainly focus on matching the low-level appearance statistics of the foreground with the background based on handcrafted heuristics, such as global and local color distribution [18], [19], [20], [21], gradient information [4], [22], [23], multiscale statistical features [24], and semantic information [25], [26]. These methods achieve image harmony by directly matching appearance statistics, but these methods cannot effectively improve the visual effect. Johnson et al. [27] addressed the lack of realism with a data-driven approach by retrieving a set of real images with similar global layouts for transferring appearances. With the development of deep learning, many deep-learning-based methods have emerged recently [16], [25], [28], [29], [30], [31], [32], [33], [34]. Most of the deep-learning-based methods are based on variants of U-net [35] as their backbone [16], [28], [28], [31], [34], [36] explored various attention mechanisms for harmonization. The work in [16] was based on the GAN [12] network to add discriminators to further improve the quality and realism of the harmonized image. Cong et al. [37] and [16] treated foreground and foreground as distinct domains and explicitly exploited the information contained in the background to guide foreground harmonization. Tsai et al. [38] and Sofiiuk et al. [25] leveraged auxiliary semantic features to improve the basic image harmonization network. Different from the existing methods, we are more concerned with the results of image harmonization for the performance of downstream object detection tasks, so we combine the advantages of [16] and [25] to further propose a task-oriented image harmonization network.

C. Physics-Based Noise Model

Image synthesis is an effective method to solve the problem of insufficient datasets, especially for remote sensing [14], [15]. To reduce the difference between synthetic images and real images, recent research introduces sensor noise into synthetic images. This approach is based on the analysis of the noise component of the detector. Noise component analysis plays a critical role in traditional sensor noise suppression [39], [40], [41]. However, the noise component of the detector cannot be directly used for image synthesis. Instead, image synthesis needs a noise model of digital image and noise parameter evaluation based on noise component [42], [43], [44], [45]. Foi et al. [42] propose an algorithm for the fully automatic estimation of the Poisson–Gaussian noise model parameters. In the field of remote sensing, Mosley et al. [46] propose a noise model for a narrow-angle camera on board the Lunar.

More recently, the noisy image synthesis approach has become popular for image denoising and reconstruction [47], [48], [49], [50]. In these approaches, neural networks are used to generate noisy data pairs. Wei et al. [51] proposed a physics-based noise formation model of CMOS and synthesized the noisy images to train the denoising network. In our work, we establish a general model for remote sensing image noise based on the noise components and optimize the model according to a specific condition.

III. Methodology

The overall procedure of our proposed RS-ISP image synthesis framework is shown in Fig. 2. The three stages of the proposed RS-ISP image synthesis framework are coarse image synthesis, image harmonization, and noise simulation. The first stage involves the construction of large-scale coarse datasets, while the second and third stages are subsequently used to alter the coarse images created in the first stage. We set up the harmonization process in the second stage and the noise simulation process in the third stage rather than the opposite

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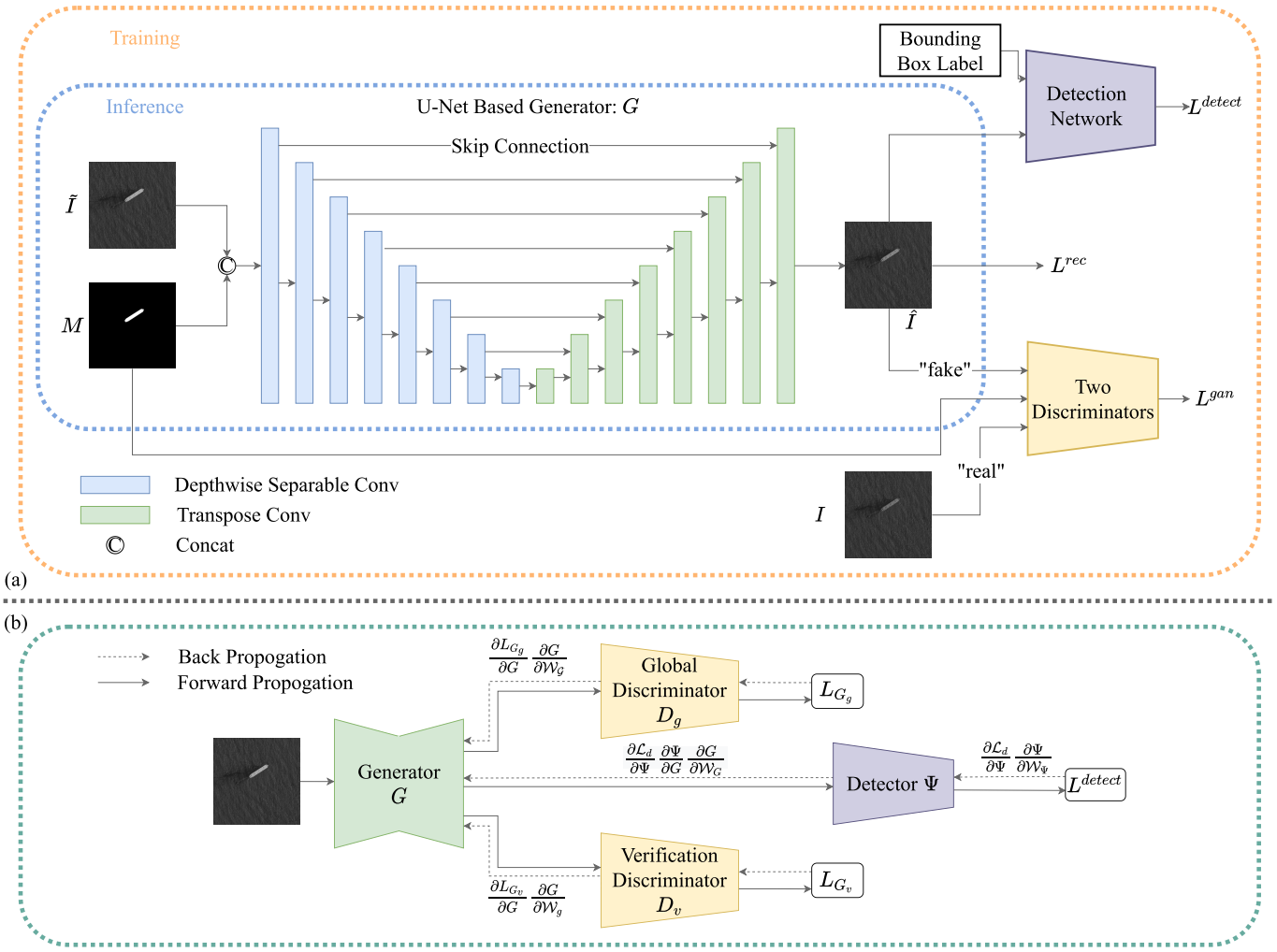


Fig. 3. Illustration of our proposed detection-oriented image harmonization network. (a) Basic structure and (b) gradient flow with respect to the generator G.

to prevent noise distribution from being modified by the harmonization process. In subsections, we describe our proposed image synthesis framework in the order of the image synthesis process in detail. Section III-A details our physically guided image synthesis method based on real remote sensing scenes. Section III-B details our improved detection-oriented image harmonization method based on DoveNet [16]. We analyze the noise model in the process of image capturing and detail our noise simulation model in Section III-C.

A. Basic Method of Image Synthesis

The generated image is mainly composed of two parts: one is the foreground which contains the object image, and the other is the background. To obtain a realistic and rich foreground image, we construct a high-quality 3-D model of the object through laser scanning and then project the 3-D model to obtain a multipose object image as the foreground area of the generated image. To ensure consistency in the distribution of synthetic and real images, we use pure ocean images from the GaoFen-2 dataset that do not contain any ship targets as background materials for synthetic images.

Finally, randomly pasting the obtained foreground image onto the background yields our coarse synthesis images. In addition, we determine the size of the foreground according to the external parameters of the sensor during background imaging, to ensure that images with unreasonable foreground sizes will not be generated. The general process is shown in Fig. 2 stage 1.

B. Detection-Oriented Image Harmonization

In the coarse synthesis image generated in the first stage, the foreground in the synthesized image is projected from the 3-D model, and the background is obtained from the real data. Hence, there is a domain gap between the foreground and the background (e.g., lighting conditions), which leads to a large distribution shift between the synthetic image and the real image. On the other hand, the integrity and structure of the semantic information of the foreground region, which is the key to the downstream object detection task, are somewhat destroyed by the incongruity of the foreground and the foreground. If this synthetic image is directly used as the training set for the downstream object detection task, the difference between the training set and the test set will make the trained

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object detection model perform poorly on the test set. We have confirmed this point through experiments.

To reduce generated images inconsistency and ensure distribution consistency between the generated images and real images, we first build a lightweight image harmonization network based on DoveNet [16]. We further construct a detection-oriented image harmonization network based on the following two reasons.

1) Our goal of using an image harmonization network is to improve the realism of synthetic images so that object detection models trained on synthetic images perform better on real data.
2) Image harmonization algorithm usually only focuses on low-level semantic information in the image and lacks the ability to reconstruct high-level semantic information, which is fatal to object detection algorithms.

The detail of our proposed detection-oriented image harmonization network is illustrated in Fig. 3.

Given a real image I, its corresponding synthesis image $\widetilde{I}$ whose foreground has been transformed with different kinds of lighting conditions and a foreground mask M, the goal of image harmonization network is to generate $\hat{I}$ with synthesized image $\widetilde{I}$, and $\hat{I}$ is expected to be as close to the real image I as possible. In our proposed detection-oriented image harmonization network, it contains an attention-enhanced U-Net generator G, two different discriminators D_g and D_v to guide G for generating more realistic and harmonious images, and a YOLOv5s model Ψ to help G generate images that meet both the visual and task goals.

Generator G is an attention-enhanced U-Net [35] with skip connections from encoders to decoders. Considering our task scenario is relatively simple and to alleviate the increase in parameters caused by the addition of YOLOv5s, we replace the convolutional layers in the generator with depthwise separable convolutions to reduce parameters as much as possible without sacrificing accuracy. Finally, we use L1 distance as the reconstruction loss to enforce generated image $\hat{I} = G(\widetilde{I}, M)$ close to the ground-truth real image I

$$L_G^{\text{rec}} = ||\hat{I} - I||_1. \tag{1}$$

Global discriminator D_g takes I as real images and $\hat{I}$ as fake images. During training, the goal of D_g is to distinguish between real and fake images as much as possible, while G is to generate $\hat{I}$ as real as possible to fool D_g. We leverage hinge loss for stabilizing training, which is given by the following equation:

$$L_{D_g} = E[\max(0, 1 - D_g(I))] + E[\max(0, 1 + D_g(\hat{I}))] \tag{2}$$

$$L_{G_g} = E[D_g(G(\widetilde{I}, M))]. \tag{3}$$

The foreground and background of a real image are captured in the same condition and thus belong to the same domain. But the foreground of the synthesized images is projected from 3-D models, which means the foreground and the background belong to different domains. Domain verification discriminator D_v is to verify whether the foreground and the background of a given image belong to the same domain. We use partial convolution to build a domain representation extractor F to extract domain representations of the foreground and the background while avoiding information leakage. Formally, given a real image I, $I_f = I \odot M$ and $I_b = I \odot (1 - M)$ represent the masked foreground region and the masked background region, respectively. Then we define $l_f = F(I_f, M)$ as the foreground representation and $l_b = F(I_b, (1 - M))$ as the background representation. Similarly, we define $\hat{l}_f, \hat{l}_b$ as the foreground and background representations of harmonized image $\hat{I}$.

After obtaining domain representations, we calculate the domain similarity

$$D_v(I) = l_f \cdot l_b \tag{4}$$

$$D_v(\hat{I}) = \hat{l}_f \cdot \hat{l}_b \tag{5}$$

and the loss function of the domain verification discriminator can be written as follows:

$$L_{D_v} = E[\max(0, 1 - D_v(I))] + E[\max(0, 1 + D_v(\hat{I}))] \tag{6}$$

$$L_{G_v} = -E[D_v(G(\widetilde{I}))]. \tag{7}$$

So far, the GAN loss for training generator G is

$$L_G^{\text{gan}} = L_{G_g} + L_{G_v}. \tag{8}$$

We have proved through experiments that the harmonized network can greatly reduce the difference between the foreground and the background and greatly improve the performance of the object detection algorithm on real data. Considering that our ultimate goal of harmonizing synthetic images is to improve the performance of the object detection models on real data, we combine the image harmonization network and the object detection network to further propose a detection-oriented image harmonization network. The output from G is directly input into the object detection network, and then the loss function of object detection is calculated. This loss function supervises the learning of the object detection network and the generator G at the same time, so the images $\hat{I}$ generated by G can be better adapted to the object detection task. In this way, we can make the images generated by G satisfy both visual inspection and computer perception.

We formulate the detection-oriented image harmonization as a bilevel optimization model

$$L^{\text{detect}} = \mathcal{L}_d(\Psi(\hat{I}; \mathcal{W}_\Psi)) \tag{9}$$

$$\hat{I} = G(\widetilde{I}, M; \mathcal{W}_G) \tag{10}$$

where $\mathcal{L}_d$ is the detection-specific training loss function and L^{detect} is the detection loss calculated by $\mathcal{L}_d$. Ψ denotes a detection network with learnable parameters $\mathcal{W}_\Psi$. We adopt YOLOv5s [17] as our backbone for the object detection network Ψ, where $\mathcal{L}_d$ is also following its setting. G is the generator in DoveNet defined earlier, and $\mathcal{W}_G$ is the learnable parameter in G.

So far, the total loss for training generator G is

$$L_G = \lambda_1 L_G^{\text{rec}} + \lambda_2 L_G^{\text{gan}} + \lambda_3 L^{\text{detect}} \tag{11}$$

where λ_1, λ_2, and λ_3 are the tradeoff parameters.

Fig. 3(b) illustrates the flow of gradient propagation to cooperatively train the harmonization and detection networks.

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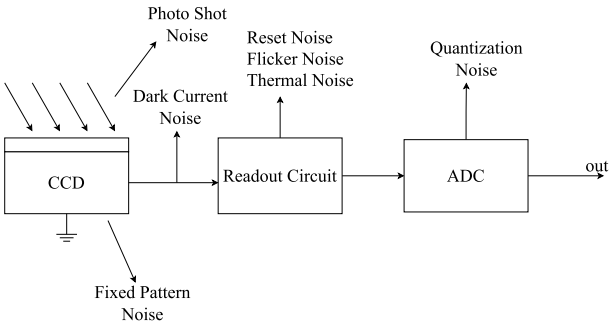


Fig. 4. Illustration of the noise components that are considered in our simulation model.

The loss gradients with respect to $\mathcal{W}_\Psi$ and $\mathcal{W}_G$ are calculated as follows:

$$\text{grad}_\Psi = \frac{\partial L^{\text{detect}}}{\partial \mathcal{W}_\Psi} = \frac{\partial \mathcal{L}_d}{\partial \Psi}\frac{\partial \Psi}{\partial \mathcal{W}_\Psi} \tag{12}$$

$$\begin{aligned}\text{grad}_G &= \lambda_1 \frac{\partial L_G^{\text{rec}}}{\partial \mathcal{W}_G} + \lambda_2 \frac{\partial L_G^{\text{gan}}}{\partial \mathcal{W}_G} + \lambda_3 \frac{\partial L^{\text{detect}}}{\partial \mathcal{W}_G} \\ &= \lambda_1 \frac{\partial L_G^{\text{rec}}}{\partial G}\frac{\partial G}{\partial \mathcal{W}_G} + \lambda_2 \frac{\partial L_G^{\text{gan}}}{\partial_G}\frac{\partial G}{\partial \mathcal{W}_G} \\ &\quad + \lambda_3 \frac{\partial \mathcal{L}_d}{\partial \Psi}\frac{\partial \Psi}{\partial G}\frac{\partial G}{\partial \mathcal{W}_G}.\end{aligned} \tag{13}$$

It is worth noting that we use the detection loss L^{detect} to supervise both the object detection network and the image harmonization network, but the effect of L^{detect} on the object detection network and the harmonization network is different. As shown in formulas (12) and (13), L^{detect} dominates the learning process of the detection network, but for the generator G in the harmonized network, since multiple losses are supervising the learning of G at the same time, we use λ_3 to make sure that L^{detect} does not play a dominant role in the learning process of G.

C. Image Synthesis With Sensor Noise

Unprocessed remote sensing images usually contain regular noise that can be modeled based on the physical properties of the sensor. To generate more realistic remote sensing images, we add sensor noise to the harmonized synthesis images. The sensor noise link contains several components, as illustrated in Fig. 4, and the sensor noise can be described by [39], [40], and [43]

$$D = K(I + N_p + N_{\text{FPN}}) + N_d + N_f + N_q \tag{14}$$

where D is the raw image detector in data numbers (DNs), K represents the sensor gain, I is the mean photon signal, N_p is the photo shot noise, N_{FPN} is the fixed pattern noise (FPN), N_d is the dark current noise, N_r is the noise generated by the readout circuit noise, and N_q is the quantization noise. In the following, we analyze each noise source and simplify the noise model.

The photo shot noise N_p is due to the inherent randomness of photons arriving at the sensor. The photo shot noise is related to the intensity of incident light and obeys

$$(I + N_p) \sim \mathcal{P}(I) \tag{15}$$

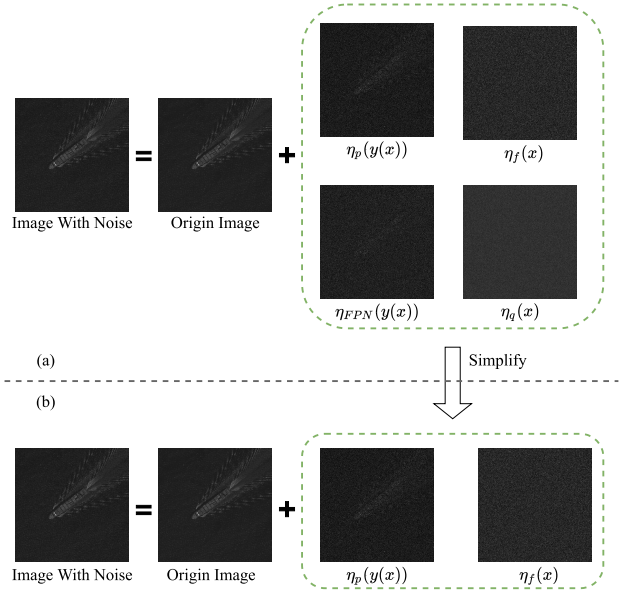


Fig. 5. Illustration of the components in the noise image. (a) Noise model including all the noise sources, η_p, η_f, η_{FPN}, and η_q, respectively. (b) Simplified noise model which removed η_{FPN} and η_q.

where $\mathcal{P}(\cdot)$ denotes the Poisson distribution. As the illumination decreases, the proportion of the photo shot noise will increase.

The dark current noise N_d originates from the output signal generated by the CCD in the state of no light and no electrical input. The basic cause of dark current is the thermal motion of electrons inside the CCD detector, and it follows the Poisson distribution:

$$(T + N_d) \sim \mathcal{P}(T) \tag{16}$$

where T is the mean number of the thermally activated charge carriers. This noise intensity is proportional to the exposure time.

FPN is a fixed position noise caused by inconsistent charge accumulation due to dark current, including dark signal nonuniformity (DSNU) and photo response nonuniformity (PRNU) [43]. DSNU is the difference in output between pixels under dark conditions, and PRNU is the difference in responsivity between pixels under uniform illumination. Through ground and on-prohibit calibration, we can establish a parametric model of FPN and determine the parameters of each pixel.

Reset noise, flicker noise, and resistor thermal noise are generated by the sensor readout circuit. The reset noise is due to the on-resistance of the reset transistor and flicker noise is low-frequency random noise caused by semiconductors. Resistor thermal noise is generated by the resistance and capacitance of the circuit, which obeys the Gaussian distribution. The combined noise of dark current noise, reset noise and amplifier noise output by the amplifier, is also known as the noise floor N_f. Which is a signal-independent noise that obeys

$$N_f \sim G(0, \sigma) \tag{17}$$

where $G(\cdot)$ denotes the Gaussian distribution. The parameter σ can be determined by testing in a dark environment.

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The quantization noise N_q is introduced during conversion from voltage to digital numbers by analog-to-digital converter (ADC). The quantization noise N_q obeys uniform distribution [51]

$$N_q \sim U\left(-\frac{1}{2q}, \frac{1}{2q}\right) \tag{18}$$

where q is the quantization step. Quantization noise has little effect under normal lighting conditions, but its effect on low-light imaging needs to be considered.

After the above analysis, we can describe the noise distribution model in the imaging process as follows:

$$\mathcal{N}(y(x)) = \eta_p(y(x)) + \eta_f + \eta_{\text{FPN}}(y(x)) + \eta_q \tag{19}$$

where $y(x)$ is value of image pixel in position x, and $\mathcal{N}(y(x))$ is the noise value of image pixel in position x. $\eta_p(y(x))$ denotes the combination of photo shot noise N_p and dark current noise N_d which follows the Poisson distribution, η_f denotes the floor noise N_f which follows the Gaussian distribution, $\eta_{\text{FPN}}(y(x))$ denotes the FPN N_{FPN} and η_q denotes the quantization noise N_q which follows uniform distribution. The Poisson noise $\eta_p(y(x))$ is characterized as follows:

$$\eta_p(y(x)) = \chi(y(x) + \eta(y(x))) \sim \mathcal{P}(\chi y(x)) \tag{20}$$

where χ is the adjustable parameter.

It is worth noting that although many noises from different sources are included in the process of the image imaging process, not all noises will seriously affect the image quality. Among several noises, the distribution of Poisson noise and Gaussian noise is random, and these two types of noise have a greater impact on image quality. The FPN can be partially determined by the testing detector, and it is easy to reduce its influence. Quantization noise is very weak and only needs to be considered when the image signal-noise ratio (SNR) is very low, such as in low-light images. Therefore, we further simplify the noise model in (19) to a combination of the Gaussian and Poisson distributions as follows:

$$\hat{\mathcal{N}}(y(x)) = \eta_p(y(x)) + \eta_f(x) \tag{21}$$

$$\hat{\mathcal{N}}(y(x)) \sim \mathcal{P}(\chi y(x)) + G(0, \sigma) \tag{22}$$

where χ and σ are adjustable parameters, which are determined by the specific sensor characteristics. The synthesis images with noise are shown in Fig. 5.

IV. Experimental

A. Implementation Details

1) Construction of Dataset: To conduct the experiments in this article, we construct two separate datasets. One is used to pretrain the image harmonization model used in the second stage of the proposed framework, and another is the synthesis dataset constructed by the proposed framework to verify the effectiveness of our method. Both the datasets are constructed based on the images from the GaoFen-2 satellite, whose local time of descending node is 10:30 A.M. The spatial resolution of images is 1 m, and they are single-channel uncompressed panchromatic images, which retain original information.

Algorithm 1 Training Strategy of the Detection-Oriented Image Harmonization Network

Input Coarse image $\widetilde{I}$, corresponding mask M, real image I and Bbox label L.

for e < max epoch **do**
  for i < iter **do**
    Random select input $\widetilde{I}_i$, M_i and label L_i;
    Input $(\widetilde{I}, M)$ into G and get output $\hat{I}$;
    Calculate reconstruction L_G^{rec};
    Input $(\hat{I}, M, I)$ into D_v and D_g and calculate adversarial loss L_{D_v} and L_{D_g};
    Update parameters in D_v and D_g;
    Input $(\hat{I}, M, I)$ into D_v and D_g and calculate adversarial loss for G by L_G^{gan};
    Input $\hat{I}$ into detection network Ψ and calculate detection loss L^{detect};
    Update parameters in detection network Ψ;
    Calculate loss function for G by $L_G = \lambda_1 L_G^{rec} + \lambda_2 L_G^{gan} + \lambda_3 L^{detect}$;
    Update parameters in generator G;
  end for
end for

TABLE I
Detection Results by Models Trained on Synthetic Datasets With and Without the Physically Simulated Noise

Real Data	F1 Score	AP@.50	AP
w/o Noise	0.203	0.148	0.051
w/ Noise	0.042	0.022	0.005

For the dataset used to train the image harmonization model, we constructed a paired dataset containing object bounding box labels based on real samples from the GaoFen-2 dataset. Specifically, we selected 323 samples with marine backgrounds and ship objects from the GaoFen-2 dataset. For each sample, we randomly selected a foreground target and manually labeled its pixel-level foreground mask. Then, four different color transformation methods mentioned in DoveNet [16] were used to process the foreground area to obtain four corresponding fake samples. The paired dataset used for training the image harmonization model was then obtained, which had a total of 1292 pairs of samples and split into a training set and validation set with a ratio of 0.9.

For the dataset used to verify the effectiveness of the proposed framework, we constructed a total of 15 000 samples following our synthesis pipeline, and then we split it into a training set and validation set with a ratio of 0.9. Besides, we use 192 real images with ship objects from the GaoFen-2 dataset as our main test set to evaluate the authenticity of synthesis images by the cross-domain test of the object detection model. Moreover, we construct two additional test sets using all the samples in HRSC2016 [52] and samples containing only ship objects in DIOR [53] to test the generalization of images generated by our framework via cross-domain testing of the object detection model.

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TABLE II
DETECTION RESULTS OF THE MODELS TRAINED ON SYNTHESIS DATA GENERATED BY OUR RS-ISP FRAMEWORK USING DIFFERENT PARAMETERS IN THE NOISE SIMULATION MODEL

-	w/o Noise			w/ Noise		
Noise Parameters in Synthetic Dataset	F1 Score	AP@.50	AP	F1 Score	AP@.50	AP
w/o Noise	0.577	0.454	0.168	0.566	0.435	0.165
$\chi = 5, \sigma = 0.03$	0.608	0.498	0.191	0.607	0.507	0.200
$\chi = 10, \sigma = 0.02$	0.595	0.450	0.167	0.581	0.445	0.160
$\chi = 20, \sigma = 0.01$	0.592	0.435	0.172	0.606	0.452	0.174

TABLE III
COMPARISON WITH OTHER IMAGE HARMONIZATION METHODS

Model	Train set	Test					
		w/ Noise			w/o Noise		
		F1 Score	AP@.50	AP	F1 Score	AP@.50	AP
DoveNet [16]	w/ Noise	0.477	0.34	0.134	0.452	0.319	0.122
	w/o Noise	0.533	0.379	0.146	0.527	0.38	0.144
S2CRNet [54]	w/ Noise	0.332	0.246	0.091	0.392	0.291	0.110
	w/o Noise	0.071	0.036	0.012	0.399	0.258	0.091
Ours	w/ Noise	0.598	0.468	0.178	0.598	0.461	0.176
	w/o Noise	0.566	0.434	0.165	0.577	0.454	0.168

TABLE IV
ABLATIVE STUDIES OF OUR PROPOSED FRAMEWORK

Index	Method				w/o Noise			w/ Noise		
	Baseline	Harmonization		Noise						
		Non-Oriented	Detection-Oriented		F1 Score	AP@.50	AP	F1 Score	AP@.50	AP
1	+	-	-	-	0.203	0.148	0.051	0.042	0.022	0.005
2	+	+	-	-	0.492	0.450	0.122	0.447	0.318	0.110
3	+	-	+	-	0.577	0.454	0.168	0.566	0.435	0.165
4	+	-	-	+	0.272	0.180	0.059	0.252	0.163	0.059
5	+	+	-	+	0.579	0.438	0.159	0.565	0.427	0.154
6	+	-	+	+	0.598	0.461	0.176	0.598	0.467	0.178

TABLE V
COMPARISON BETWEEN DETECTION-ORIENTED IMAGE HARMONIZATION NETWORK AND NON-ORIENTED IMAGE HARMONIZATION NETWORK

Model	MSE	PSNR	SSIM
Non-Oriented	16.4	35.65	0.972
Detection-Oriented	4.35	41.51	0.983

2) Training Strategy and Hyperparameters of Network: In our proposed image harmonization network, a generator G, two discriminators D_v, D_g, and an object detection network Ψ are included to guide the learning of the generator G. Among them, the two discriminators D_v and D_g improved the ability of the generator through adversarial learning with the generator, and the object detection network Ψ allows the generator to have a better perception of the needs of downstream tasks through joint learning. Specifically, the discriminators and the generator are updated alternately during the training process, while the generator and the object detection network are updated simultaneously. The training strategy of the proposed detection-oriented image harmonization model is summarized in Algorithm 1.

For the hyperparameters of the pretrain process of the proposed detection-oriented image harmonization network, we set tradeoff parameters $\lambda_1 = 100, \lambda_2 = 1, \lambda_3 = 1e-4$ in L_G. And we use the Adam optimizer with a learning rate of 0.0002 to update parameters in G, D_g, and D_v. For the detection network used in the image harmonization network, we use the stochastic gradient descent (SGD) optimizer with a learning rate of 0.01 to update parameters in Ψ. The size of inputs is 1024×1024 both in the training and inference stages. We train the detection-oriented image harmonization network for 200 epochs with a batch size of 4. For other harmonization methods in the comparative experiments, we unify the batch size of 4 during training and keep the original settings of other hyperparameters.

For the object detection algorithms YOLOv5 [17], Faster-RCNN [55], and CenterNet [56] used for cross-domain testing,

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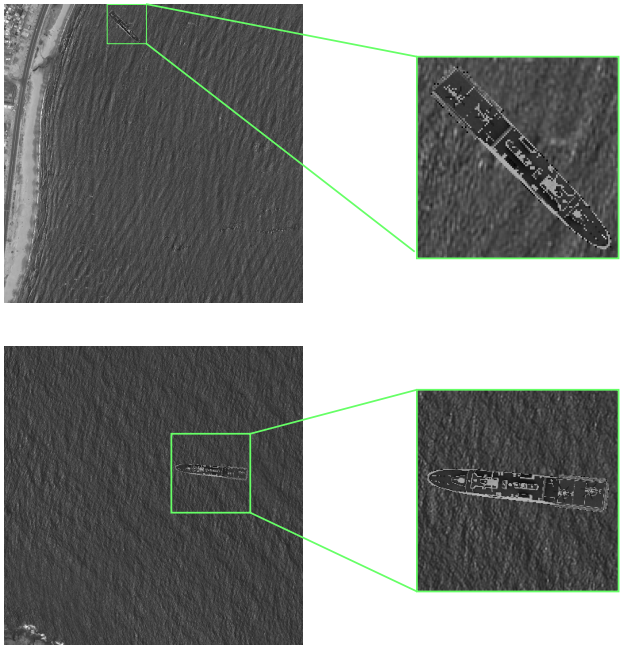


Fig. 6. Illustration of the coarse synthesis image. It can be clearly seen that obvious inconsistency exists by directly pasting the projection of the target 3-D models onto the backgrounds of the GaoFen-2 dataset without any modification.

we keep their hyperparameters consistent when training on different synthesis datasets and remove the data augmentation operations to avoid their impact on synthetic data. Among them, FasterRCNN and CenterNet use the implementation of MMDetection [57]. Besides, we set $(\chi = 5, \sigma = 0.03)$, $(\chi = 10, \sigma = 0.02)$, and $(\chi = 20, \sigma = 0.01)$ three combinations in the noise simulation stage to test the effect of noise.

B. Results of Object Detection Trained on Basic Synthesis Images

To make our proposed detection-oriented image harmonization network and sensor noise model have a basic reference, we use the coarse synthetic dataset which has not been harmonized and no extra noise added to train the detection network and test it on the real dataset with or without noise added. We use F1 score, AP@.50, and AP as the evaluation indicators of the detection results. The cross-domain experimental results in Table I show that the object detection model trained on coarse synthetic images performs extremely poorly on real data. This indirectly proves that there is a very large distribution shift and domain gap between the coarse synthetic data and real data. Our coarse synthetic image is shown in Fig. 6, the edge of the foreground in the coarse synthetic image has obvious synthetic traces, and there is a very obvious incongruity between the foreground and the background.

C. Results of Object Detection Trained on Synthesis Images Followed Our RS-ISP Framework

In this section, we conduct cross-domain experiments to evaluate the quality of the dataset synthesized using the RS-ISP framework. Table II and Fig. 9 shows the performance of the object detection model which is trained on synthetic data generated by our RS-ISP framework on real data. We add different degrees of noise on real data according to our noise model to test the performance of the model on real data containing noise. The setting of the noise parameters is empirical, and the effect of different parameters' settings $(\chi = 20, \sigma = 0.01)$, $(\chi = 10, \sigma = 0.02)$, $(\chi = 5, \sigma = 0.03)$ on image quality degradation is incremental. The column of "w/ Noise" in Table II shows the average results of three test sets with the above three noise parameters. The noise parameters are used to test the performance of synthetic images with noise consistent with the test dataset. In practical applications, the noise parameters are obtained through fitting calculations or camera tests to adjust the distribution of noise on the synthesized images. The results in Table II reveal a trend whereby better outcomes are obtained when the training set's noise level is higher (noise has a greater impact on image quality) than the test set's.

Compared with the results in Table I, the performance of the YOLOv5s model test on real data is significantly improved when it is trained on synthesis data generated by our RS-ISP framework. The comparison of the cross-domain test results of the YOLOv5s model in Tables I and II indirectly proves that the distribution shift between synthetic data and real data can be greatly reduced by our detection-oriented image harmonization method and noise simulation model.

D. Comparative Study With Other Image Harmonization Methods

In this section, we compare our detection-orientation image harmonization with two additional image harmonization methods, which are DoveNet [16] and S2CRNet [54]. To demonstrate the superiority of the proposed detection-oriented image harmonization method, we use DoveNet, and S2CRNet as the image harmonization method in the second stage of the proposed RS-ISP framework while leaving the other two stages unchanged. We also assess synthetic image quality with the same cross-domain object detection test. The comparison results are shown in Table III. The experimental results reveal that our method surpasses previous harmonization methods by a wide margin, demonstrating the superiority of our detection-oriented image harmonization method. In addition, compared with the baselines in Table I, using other image harmonization methods also achieves significant improvement, which also illustrates the rationality and flexibility of our proposed three-stage image synthesis framework.

In addition, we compare the visualization results of the aforementioned harmonization methods, which are displayed in Fig. 8. Because S2CRNet is a curve rendering method, it does not have the foreground reconstruction process, and hence its harmonizing output preserves the most details. However, DoveNet and ours are not perfect in the preservation of details in the process of reconstructing the foreground.

E. Visual Comparison of Synthetic and Real Images

As illustrated in Fig. 7, we select some samples of synthetic images generated by our RS-ISP method and real images from

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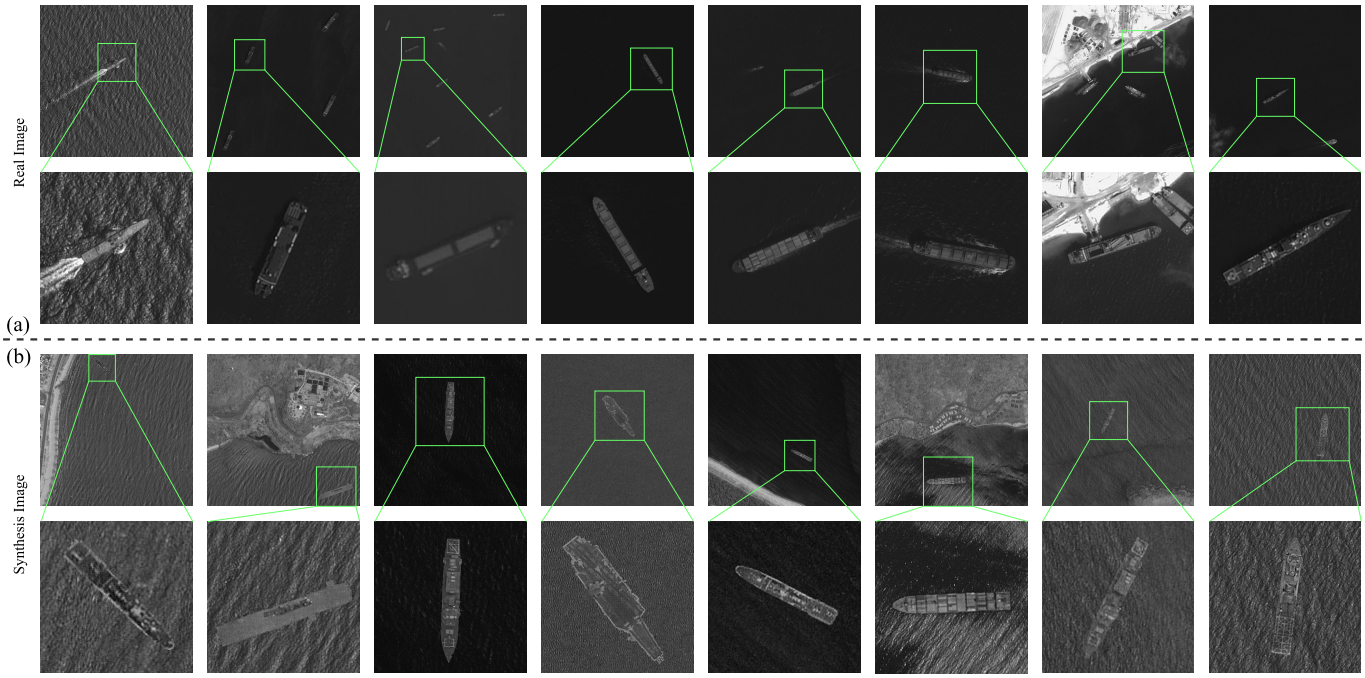


Fig. 7. Visual comparison between synthetic images generated by RS-ISP and real images. (a) Real images from GaoFen-2 dataset. (b) Synthesis images generated by RS-ISP.

TABLE VI
GENERALIZATION ABILITY EVALUATION USING FASTERRCNN AND CENTERNET AS DETECTION MODEL

Model	Setting	w/o Noise		w/ Noise	
		AP@.50	AP	AP@.50	AP
FasterRCNN	Coarse	0.103	0.036	0.103	0.038
	Coarse + H	0.642	0.320	0.596	0.273
	Coarse + H + N	0.680	0.345	0.666	0.334
CenterNet	Coarse	0.163	0.059	0.105	0.025
	Coarse + H	0.263	0.081	0.173	0.050
	Coarse + H + N	0.281	0.086	0.298	0.009

TABLE VII
GENERALIZATION TESTING ON PUBLIC DATASETS

Training set	HRSC2016 [52]			DIOR [53]		
	F1 Score	AP@.50	AP	F1 Score	AP@.50	AP
Baseline	0.024	0.006	0.001	0.054	0.024	0.006
RS-ISP	0.089	0.032	0.009	0.04	0.026	0.009

the GaoFen-2 dataset for visual comparison in this section. Overall, the synthetic image is fairly similar to the real image in terms of visuals, but it is still missing some information about how the ship interacts with its surroundings, such as the ship's shadow on the sea and the tail's track when it is sailing, which is difficult to simulate by copy–paste-based method.

F. Ablation Studies

1) Analysis of the Effect of Image Harmonization: We discuss the effect of the image harmonization network on synthetic data through cross-domain test results of the object detection model. To verify the role of the image harmonization network, we removed the harmonization stage from our image synthesis framework, and thus the whole process of image synthesis becomes coarse image synthesis with noise simulation. The result is shown in row 4 in Table IV, compared with rows 5 and 6 in Table IV, after removing the harmonization network, the AP@.50 value of the cross-domain test result of the YOLOv5s model on real data with noise decreases from 0.467 to 0.163, and the result on real data without noise decreases from 0.461 to 0.180. We can draw the conclusion from the comparison that the significant decrease in the cross-domain test result indicates that the image harmonious network can greatly improve the authenticity of the synthetic data and narrow the distribution shift between the synthetic data and the real data. The visualization results of the effect of the harmonization network on the coarse synthesis images are shown in Fig. 8, and line (A) is the coarse synthesis image and lines (B) and (C) are the harmonized images by the non-oriented harmonization network and detection-oriented harmonization network. It is obvious that in the coarse synthesis images, the ships in the foreground area appear to be floating in the air, which is incompatible with the background, whereas the harmonized images are much better and appear to be more harmonious.

2) Analysis on the Effect of the Detection-Oriented in Image Harmonization: To make the results of image harmonization better applicable to object detection algorithms, we design a detection-oriented image harmonization network. We discuss the effect of task orientation on image harmonization results in this section. We removed the detection network used for guidance in the image harmonization network and only trained a basic image harmonization network to harmonize our coarse

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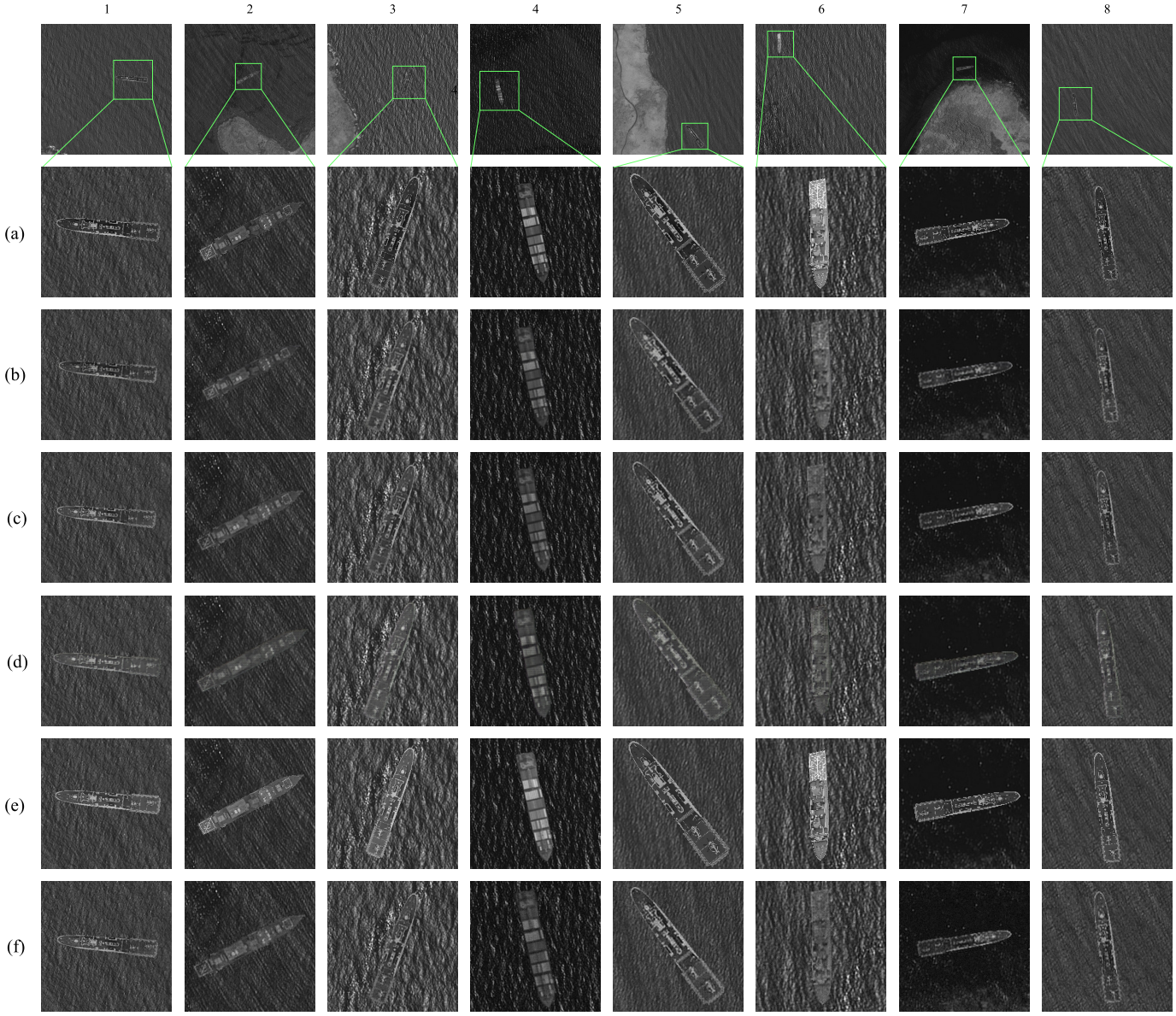


Fig. 8. Visual comparison of the synthetic images generated under different settings. (a) Coarse synthetic image. (b) Images harmonized by the non-oriented harmonization method. (c) Images harmonized by our detection-oriented harmonization method. (d) Images harmonized by DoveNet. (e) Images harmonized by S2CRNet. (f) Results by adding the physically simulated noise to (c). The difference between (b), (c), and (f) clearly shows that following the harmonization approach, the tone of the foreground is closer to the background, and the overall appearance is more consistent. Apart from the visual difference between (b), (c), and (f), the effect of our proposed RS-ISP is better demonstrated by the quantitative comparison of detection results.

synthesis images. The results of object detection after data harmonization using a non-detection-oriented image harmonization network are presented in rows 2 and 5 in Table IV. Compared with rows 3 and 6 in Table IV, it can be found that after removing the guidance, the cross-domain test result of the object detection model has dropped significantly, from 0.454 to 0.450 and 0.461 to 0.438, and the drop was more obvious on the test set containing noise, from 0.435 to 0.318 and 0.467 to 0.427. We further compared the visualization results of the two methods for the harmonization of coarse synthetic images. As shown in (B) and (C) in Fig. 8, the detection-oriented image harmonization network output (C) results in better and similar lighting conditions in the foreground and background compared with the non-oriented harmonization network output. Besides, we use the commonly used image quality evaluation metrics mean squared error (MSE), peak signal-to-noise ratio (PSNR), and structure similarity (SSIM) to compare the output results of the two networks. As shown in Table V, after adding the object detection YOLOv5s model to guide the image harmonization process, the images generated by the network have a significant improvement in PSNR and SSIM, and the MSE is also significantly reduced.

We believe that the reason why the image harmonization network can generate more realistic images after adding detection guidance is that the basic harmonization network only pays attention to the similarity at the pixel level in the process of image reconstruction and seriously ignores the semantic integrity of the reconstructed image. However, object detection

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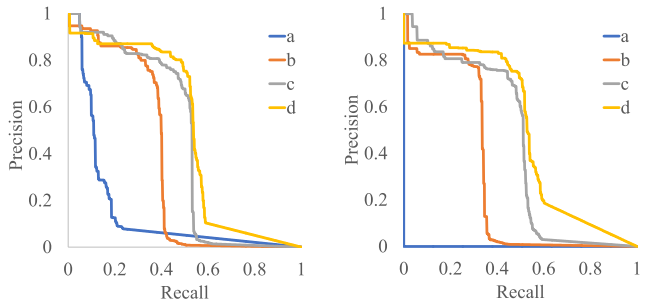


Fig. 9. PR curves for detection results on real datasets by testing the YOLOv5s model trained on synthetic datasets generated by different settings (represented by different colored lines). (a) Testing results on the real dataset without adding physically simulated noise. (b) Testing results on the real dataset with noise parameters ($\chi = 5, \sigma = 0.03$). Line a is the coarse synthetic dataset, line b is the coarse synthetic dataset harmonized by the non-oriented harmonization network, line c is the coarse synthetic dataset processed by our detection-oriented harmonization network, and line d is the synthetic dataset generated according to our proposed RS-ISP framework.

needs to use high-level semantic information to complete, so it can provide guidance for the results of image harmonization from the semantic level during the training process, thus making up for the lack of image harmonization network's ability to reconstruct semantic information.

3) Analysis on the Effect of the Noise Model: Since synthetic data are acquired differently from real images, synthetic data typically contain no noise or only a small amount of noise. This difference in noise can further enlarge the distribution shift of synthetic and real data. Thus, we discuss the role of noise simulation in this section. To verify the effect of noise on the distribution of synthetic data, we removed the noise simulation stage from our image synthesis framework and directly used the harmonized results to train and test the object detection model. The results are shown in rows 1–3 in Table IV. Compared with rows 4–6 in Table IV, after adding noise, the cross-domain test results of the object detection model on real data can be further improved, and the improvement is more obvious when the test set contains noise. This phenomenon proves that noise can further reduce the distribution difference between synthetic data and real data, making synthetic data closer to real data. Besides, comparing results of rows 1 and 4 in Table IV shows that even adding noise directly to the coarse synthesis images can significantly increase the cross-domain test results of the object detection model, which indicates that the contribution of our noise model to the similarity of the distribution of the synthetic data and real data is independent of image harmonization.

4) Generalization Test on Synthetic Data: In this section, we further discuss the generalization performance of the synthetic data generated by RS-ISP to other object detection models. We use two well-known object detection models, FasterRCNN and CenterNet, and run the same cross-domain test experiments to test synthetic data generalizability. Generalization test results on FasterRCNN and CenterNet are shown in Table VI. Specifically, we train the object detection models on synthetic datasets under three different conditions and evaluate them on real data. The three settings in Table VI are coarse synthetic dataset (Coarse), harmonized coarse synthesis dataset (Coarse + H), and harmonized coarse synthesis dataset with noise simulation (Coarse + H + N). The generalization test results in Table VI show that the synthetic data generated by RS-ISP work well for a variety of object detection models and that the synthetic images generated by RS-ISP are highly similar to real image distribution.

G. Discussion and Limitations

In this section, we discuss some limitations of this work. To determine whether the datasets created by RS-ISP work well on datasets with a wide range of distributions, we investigate its effectiveness on two publicly available datasets HRSC2016 [52] and DIOR [53]. Specifically, we still use the synthetic data constructed based on GaoFen-2 as the training set and validation set and use two public datasets as the test set to conduct the same experiments as before. Table VII shows the test results for HRSC2016 and DIOR. For "Baseline," coarse synthetic data were used as the training set, and for "RS-ISP," the proposed RS-ISP framework was used to produce the training data. Since the samples in HRSC2016 and DIOR and the samples in GaoFen-2 are derived from different satellite sensors, the domain differences between datasets are relatively high. Since the image synthesized by our framework is based on the coarse synthesized image, the distribution of the synthesized image and the target dataset is further narrowed according to the domain prior information provided by the background and the sensor noise prior information. Therefore, the synthesis data construction based on the background of GaoFen-2 have a great gap between the other datasets, which in turn leads to poor results as reported in Table VII. However, a comparison of "Baseline" and "RS-ISP" in Table VII shows that although the coarse dataset built based on the background of GaoFen-2 is very different from HRSC2016 and DIOR, after processing stages 2 and 3 in our proposed framework, some slight improvements can be obtained.

V. Conclusion

In this article, we propose the RS-ISP to address the issue of insufficient data in remote sensing scenarios. RS-ISP can produce synthetic images that are similar enough to real images to train object detection models. RS-ISP first uses copy–paste to create a coarse synthetic image with diverse and adjustable targets and then uses a deep-learning-based image harmonization approach to constrain the lighting conditions of the foreground and the background to be the same. In addition, we investigate the sources of noise in the sensor imaging process and reduce them into a noise distribution model that is incorporated into the synthetic images to improve its realism. The cross-domain experimental results of multiple object detection models on the GaoFen-2 dataset show that our proposed detection-oriented image harmonization network and noise distribution pipeline can greatly improve the authenticity of synthetic images and reduce the distribution differences between synthetic images and real images.

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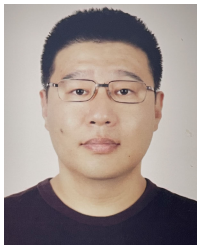
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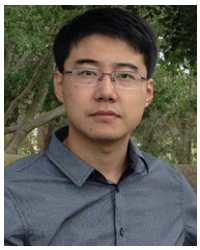
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